AP BIOLOGY TOPIC 7.7

Common Ancestry Across All Domains & Eukaryotes

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-2.A | EVO-2.B | LUCA | EUKARYOTIC SYNAPOMORPHIES

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.7 (COMMON ANCESTRY ACROSS ALL DOMAINS & EUKARYOTES)

CED Core Vocabulary

1. DNA & RNA as Genetic Material — Double-stranded DNA stores hereditary information and RNA transcripts express it via base-pairing (A-T/U, G-C).

2. Nearly Universal Genetic Code — The same 64 mRNA codons specify the exact same 20 amino acids from bacteria to humans (enabling transgenic biotechnology, e.g., bacteria expressing human insulin!).

4. Core Metabolic Pathways (Glycolysis) — Cytosolic glycolysis converts glucose to pyruvate without O_2 across virtually all organisms, showing it evolved very early before Earth had an oxygenated atmosphere.

6. Phospholipid Bilayer Cell Membranes — Selectively permeable amphipathic lipid membranes enclose all cells.

1. Membrane-Bound Organelles & Nucleus — All eukaryotes possess internal membrane-bound compartments (nucleus, ER, Golgi apparatus, lysosomes, peroxisomes) that compartmentalize incompatible cellular...

Mitochondria — are present in all eukaryotic lineages (or their vestigial remnants, mitosomes/hydrogenosomes), reflecting a single ancient endosymbiotic uptake of an aerobic...

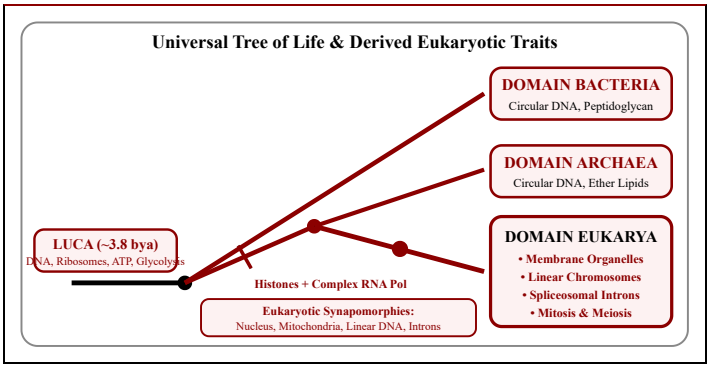
§2. UNIVERSAL COMMON ANCESTRY (LUCA): ALL LIFE ON EARTH

Bacteria, Archaea, Eukarya

- Structural and functional evidence conserved across **all three domains of life (Bacteria, Archaea, and Eukarya)** demonstrates that all living organisms descended from a single **Last Universal Common Ancestor (LUCA)**:
- **6 Universal Molecular Homologies (Shared by ALL Life):**
  - **1. DNA & RNA as Genetic Material:** Double-stranded DNA stores hereditary information and RNA transcripts express it via base-pairing (**A-T/U, G-C**).
  - **2. Nearly Universal Genetic Code:** The same 64 mRNA codons specify the exact same 20 amino acids from bacteria to humans (enabling transgenic biotechnology, e.g., bacteria expressing human insulin!).
  - **3. Ribosomes (rRNA + proteins):** All cells translate mRNA into polypeptides using ribosomes.
  - **4. Core Metabolic Pathways (Glycolysis):** Cytosolic glycolysis converts glucose to pyruvate without **O_2** across virtually all organisms, showing it evolved very early before Earth had an oxygenated atmosphere.
  - **5. ATP (Adenosine Triphosphate) & Chemiosmosis:** Universal cellular energy currency and proton-gradient ATP synthase machinery.
  - **6. Phospholipid Bilayer Cell Membranes:** Selectively permeable amphipathic lipid membranes enclose all cells.

§4. VISUALIZING THE 3 DOMAINS & EUKARYOTIC SYNAPOMORPHIES

Phylogenetic Tree of Life



§5. COMPARISON: BACTERIA VS. ARCHAEA VS. EUKARYA

High-Yield Summary Table

CELLULAR FEATURE	DOMAIN BACTERIA	DOMAIN ARCHAEA	DOMAIN EUKARYA
Nuclear Envelope	Absent (Nucleoid)	Absent (Nucleoid)	**Present (Double membrane)**
Membrane Organelles	Absent	Absent	**Present (Mitochondria, ER, Golgi)**
Chromosome Structure	**1 Circular** DNA	**1 Circular** DNA	**Multiple Linear** Chromosomes
Histone Proteins	Absent	Present (in many species)	**Present (Nucleosomes)**
Introns in Genes	Rare / Absent	Present in some genes	**Widespread (Spliceosomes)**
Cell Division	Binary Fission	Binary Fission	**Mitosis & Meiosis** (Cytoskeleton spindle)

[AP EXAM FRQ SCORING STRATEGY] **Phylogenetic Insight:** Notice that **Archaea and Eukarya are sister clades** (more closely related to each other than either is to Bacteria), evidenced by shared histones, TATA boxes, and multiple RNA polymerases!

§3. FUNDAMENTAL EVIDENCE FOR COMMON ANCESTRY OF EUKARYOTES

CED EVO-2.A Focus

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Fundamental Evidence for Common Ancestry of Eukaryotes ➔
single common eukaryotic ancestor (LECA) ➔
1. Membrane-Bound Organelles & Nucleus

- All eukaryotes (*protists, fungi, plants, and animals*) share specific derived structural and molecular features not found in Bacteria, proving inheritance from a **single common eukaryotic ancestor (LECA)**:
- **1. Membrane-Bound Organelles & Nucleus:**
  - All eukaryotes possess internal membrane-bound compartments (**nucleus, ER, Golgi apparatus, lysosomes, peroxisomes**) that compartmentalize incompatible cellular processes and increase surface-area-to-volume efficiency.
  - **Mitochondria** are present in all eukaryotic lineages (or their vestigial remnants, *mitosomes/hydrogenosomes*), reflecting a single ancient **endosymbiotic uptake** of an aerobic alpha-proteobacterium.
- **2. Multiple Linear Chromosomes + Histone Proteins:**
  - Eukaryotic nuclear DNA is organized into **multiple linear chromosomes** wrapped around **histone spools (nucleosomes)** and capped by telomeres, enclosed within a double nuclear envelope.
  - *Contrast:* Prokaryotes typically have a **single circular chromosome** located in an unbound cytoplasmic **nucleoid region**.
- **3. Genes Containing Introns & Spliceosomal RNA Processing:**
  - Eukaryotic protein-coding genes are interrupted by non-coding **introns** interspersed with coding **exons**.
  - Pre-mRNA undergoes **5′ GTP capping, 3′ poly-A tailing, and spliceosome-mediated intron removal** (enabling alternative splicing) inside the nucleus before translation.

[CORE AXIOM / REMEMBER] **Shared mitochondria = evidence of a shared eukaryotic lineage.** Mitochondria evolved *before* the endosymbiosis of chloroplasts (which occurred later only in the photosynthetic plant/algal lineage).

[MECHANISTIC NOTE] **Why Introns & Linear Chromosomes Matter:** Both are major structural synapomorphies shared across diverse eukaryotic kingdoms, strongly supporting a single monophyletic origin of Domain Eukarya.

[COMMON AP EXAM FRQ TRAP] **Do not confuse introns with exons! Introns** (*in*tervening sequences) are **removed/spliced out**; **Exons** (*ex*pressed sequences) **remain and exit** the nucleus to be translated into protein!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.7 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.7: Common Ancestry Across All Domains & Eukaryotes ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔
Experimental Perturbation (±2 SE_x̄) ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + $2pq$ (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate $2pq$ (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.7 to prove why the claim holds.

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